Manthan Chawda

+91-8780082654 | manthanchawda1212@gmail.com | linkedin.com/in/manthan-chawda-521522352 | github.com/XDecoder12

Education

GSFC University — Vadodara, Gujarat
B.Tech in Computer Science & Engineering (AI/ML Specialization) | CGPA: 8.47 — *July 2024 – Present*

Achievements

First Runner-Up – ELCIA Next-Gen Innovative Tech Hackathon 2026
Smart City Drone-AI Challenge – Team CivicPulse, among 183 teams from 18 states and 100+ institutions.
Top 20 – TetraTHON 2026 – *Indo-French AI Innovation Sprint*, Team InvoiceGuard AI.

Certifications

TIDES Ignition Program 2026 – IIT Roorkee — *January 2026*
Oracle Cloud Infrastructure Data Science Professional – Oracle — *October 2025*
Oracle Cloud Infrastructure AI Foundations – Oracle — *October 2025*
AWS Cloud Foundations – Amazon Web Services — *July 2025*

Experience

Academic Associate — January 2026 – Present
GSFC University — *Vadodara, Gujarat*
- Guide undergraduate students through experimental physics laboratory sessions and help bridge the gap between theory and practical application.
- Set up and maintain laboratory equipment and supervise hands-on experiments while ensuring proper equipment handling and laboratory safety.

Intern — December 2025
GUIITAR Council, GSFC University — *Vadodara, Gujarat*
- Researched aerodynamic and modular drone designs inspired by mini-FPV platforms with a focus on improving flight efficiency and range.
- Analyzed and proposed structural design iterations for lightweight 3D-printed wings and body components to improve aerodynamics and flight duration.

Projects

CivicPulse | *YOLO11m-seg, ByteTrack, MiDaS, Python, PyTorch, FastAPI* — August 2026 – Present
- Developed an AI-powered civic infrastructure monitoring platform that analyses drone footage to detect potholes, waterlogging, open manholes, drainage overflow and damaged footpaths.
- Built a computer vision and incident-analysis pipeline using YOLO11m-seg, ByteTrack and MiDaS and integrated it with FastAPI, React/Vite and Supabase/PostgreSQL for telemetry-aware dashboard monitoring.

ClearContract | *Gemini API, React, Chrome Extension, Node.js, Express, MongoDB* — July 2026 – Present
- Developed a Chrome extension prototype that analyses pasted contract and Terms of Service text using Google's Gemini API to identify potentially concerning clauses and explain them in simpler language.
- Built the extension-to-backend workflow using Node.js and Express with MongoDB-based application components and authentication while continuing to develop the dashboard.

InvoiceGuard AI | *Python, FastAPI, React, Gemini Vision, SQLite, Docker* — August 2026
- Developed an AI-powered invoice auditing platform for automated invoice extraction, vendor validation, anomaly detection and financial risk assessment.
- Contributed to the full-stack architecture spanning FastAPI services, React frontend, database design, AI processing and Docker-based deployment as part of a 4-member team.
- **Selected among the Top 20** teams at TetraTHON 2026, a 36-hour Indo-French AI Innovation Sprint organized by Navrachana University and ISEN Méditerranée.

Meshmerise | *C++, Arduino IDE, STM32, RLS08, TB6612FNG* — October 2025 – December 2025
- Designed and built an autonomous maze-solving robot using an STM32 microcontroller and RLS08 line sensors for navigation.
- Implemented line-following and left-hand maze-solving logic in Arduino IDE and optimized second-run traversal using recorded turn sequences.

Technical Skills

Languages: Python, C/C++, SQL, Swift, HTML/CSS, Javascript/Typescript
AI/ML: PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, NumPy, pandas, Matplotlib
Frameworks: FastAPI, React, Node.js, Express
Databases & Tools: PostgreSQL, MongoDB, Git, Github, Docker, VS Code, Arduino IDE, LaTeX